SIDEQUEST

A Student Social & Project Collaboration Platform

Project Proposal | Author: 4th Year Computer Science Engineering Student | Domain: Full-Stack System Architecture

1. EXECUTIVE SUMMARY

SIDEQUEST is a student-first social and collaboration platform designed to unify the academic, project, and social experiences of college students. Instead of distributing student interactions across multiple external tools, SIDEQUEST brings these experiences into a single ecosystem built around dynamic student profiles, structured project recruitment, and verifiable technical growth.

2. THE PROBLEM

College students currently manage their academic, project, and career activities across fragmented platforms:

- **Fragmented Communication:** Communication and group planning are split across WhatsApp, Telegram, and Discord.
- **Scattered Proof of Work:** Technical accomplishments and build history are buried across LinkedIn and GitHub.
- **In-Efficient Matchmaking:** Finding qualified project teammates with complementary technical stacks remains difficult.
- **Unstructured Opportunities:** Campus events, hackathons, and open-source opportunities are scattered across unstructured group chats.

3. PRODUCT VISION

To build a clean digital layer for campus life where students discover peers, assemble project teams, participate in communities, and track technical skill development through verified build history.

4. TARGET USERS

- **Students:** Discover peers, join project teams, and build a verified technical profile.
- **Project Leads:** Recruit teammates based on specific role requirements.
- **Campus Clubs:** Coordinate initiatives and publish events.

5. CORE PRODUCT MODULES

5.1 Profile Cards

Dynamic digital profiles representing major, year, active project status, tech stack, and verified contribution history.

5.2 Skill Ranks & Progressive Verification

Skills are evaluated across objective tiers based on practical build output rather than self-reported lists:

Bronze: Foundational knowledge of computer science concepts and syntax.

Silver: Competent practitioner with completed baseline coursework and projects.

Gold: Proven contributor who has built and deployed functional full-stack applications.

Platinum: Advanced engineer with multi-project deployments and mentorship experience.

Diamond: Lead architect with significant production or open-source impact.

5.3 Visual Skill Paths

Technical competencies represented as connected node graphs (e.g., CS Fundamentals → Java → Spring Boot → Microservices), detailing verified milestones along a structured learning pathway.

5.4 Project Matchmaking & Role Board

Project leads post project listings with specific role openings (e.g., Frontend Developer, Backend Developer). Students search listings by required skills and apply directly using their Profile Cards.

5.5 Campus Communities & Events

Structured spaces for department groups, coding clubs, and campus events with registration tracking and announcements.

6. TECHNICAL ARCHITECTURE

Layer	Technology	Function
Frontend	Next.js 14, React, TypeScript, Tailwind CSS	Renders user interface, skill paths, and profile cards.
Backend API	Java 17, Spring Boot, Spring Security	Handles REST APIs, authentication, and core business logic.
Database	PostgreSQL	Relational storage for profiles, skills, and project listings.
Caching & Auth	Redis	JWT revocation, rate limiting, and session caching.
Infrastructure	Docker, AWS S3, Vercel / Render	Containerized service hosting and cloud object storage.

7. DEVELOPMENT ROADMAP (PHASE 1 MVP)

Weeks 1 - 2: PostgreSQL database schema design, Spring Boot backend setup, and JWT authentication implementation.

Week 3: Next.js frontend setup, Profile Card UI components, and SVG Skill Path interfaces.

Weeks 4 - 5: REST APIs for project listings, skill-requirement filtering, and role application logic.

Week 6: Docker containerization, cloud deployment, and beta testing with student cohorts.

8. STRATEGIC VALUE & ENGINEERING PORTFOLIO IMPACT

Executing SIDEQUEST demonstrates proficiency in full-stack architecture, including enterprise Java Spring Boot patterns, relational database normalization in PostgreSQL, high-speed caching strategies with Redis, and modern frontend development with Next.js and TypeScript.